

High-Throughput Multiplexed Quantification of Molecules by Aptamer Sequencing (Apt-seq) in Single Cells

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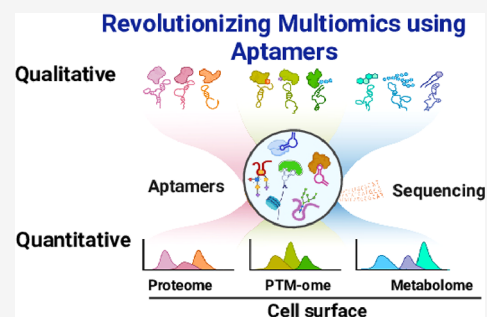


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ABSTRACT: The advent of high-throughput sequencing technologies is transforming life sciences into a quantitative paradigm. However, existing sequencing technologies for nondirectly sequencable molecules, such as antibody-sequencing and glycan-sequencing, face certain challenges, including the complexity of antibody-oligonucleotide conjugation procedures, potential steric hindrance, and the labor-intensive chemoenzymatic labeling. Aptamers, as directly sequencable nucleic acids, offer exceptional specificity, broad target range, and small size, making them ideal tools for multimodal molecular profiling. In this work, we present Apt-seq, an aptamer-based, high-throughput platform for multimodal omics quantification at single-cell resolution. This integrative strategy—termed Aptomics—enables parallel profiling of cell surface proteins, glycans, and mRNA. The feasibility of the platform was validated using commercial cell lines, demonstrating strong concordance between sequencing data and flow cytometry results at both bulk and single-cell levels. When applied to complex clinical samples, Apt-seq exhibited exceptional sensitivity and enabled the precise profiling of tumor heterogeneity. Notably, the platform identified a subpopulation of tumor cells with elevated PTK7 surface expression, a marker of stemness. These findings underscore the role of PTK7 in tumor stemness and its potential as a stem-like biomarker. Moreover, using an aptamer targeting sialic acid, we tracked the dynamic changes in sialylation during T cell differentiation, observing an increase in sialic acid levels as resting T cells transitioned into functional T cells, followed by a subsequent decline upon maturity. Collectively, we have developed a new form of multiomics, termed Aptomics, which employs aptamers to enable the sequencing of molecules that are not directly sequenceable, such as proteins and glycans, etc. Apt-seq enables the simultaneous quantitation of both directly and nondirectly sequencable molecules, offering a versatile and scalable platform for the comprehensive quantification of molecules of life in complex biological systems and advancing quantitative science.



INTRODUCTION

Life sciences are undergoing a paradigm shift from qualitative descriptions to quantitative analyses, demanding unprecedented precision in high-throughput quantification of functional molecules in living systems. The rapid advancement of high-throughput sequencing technologies has empowered researchers to accurately measure and analyze the expression and dynamic behavior of nucleic acid molecules, such as DNA and RNA, across multiple biological scales.^{1,2} These technological breakthroughs have revolutionized the study of gene expression in thousands to millions of cells, offering unparalleled insights into the intricate mechanisms governing complex biological processes. Despite these advances, existing sequencing methods predominantly focus on intracellular nucleic acids, leaving a significant gap in the scalable profiling of critical functional molecules that cannot be directly sequenced, such as proteins and small molecule metabolites like sialic acids. This limitation is particularly pronounced for cell membrane molecules, which form the crucial interface between cells and their environment.^{3,4} These molecules

include cell membrane proteins, which are central to cellular function and pathology and are the targets of over 50% of FDA-approved drugs,⁵ and cell surface glycans, which are vital for cell–cell recognition and immune modulation.^{6–10} Therefore, the simultaneous high-throughput quantification of both directly and nonsequencable entities at the single-cell level is essential.

Bottom-up mass spectrometry (MS)-based methods are essential tools for analyzing proteins and metabolites. However, challenges still persist with the complexity of sample preparation and the difficulty in integrating these methods with other omics technologies.¹¹ Currently, antibody-oligonucleotide techniques (antibody-seq) have emerged to quantify cell-

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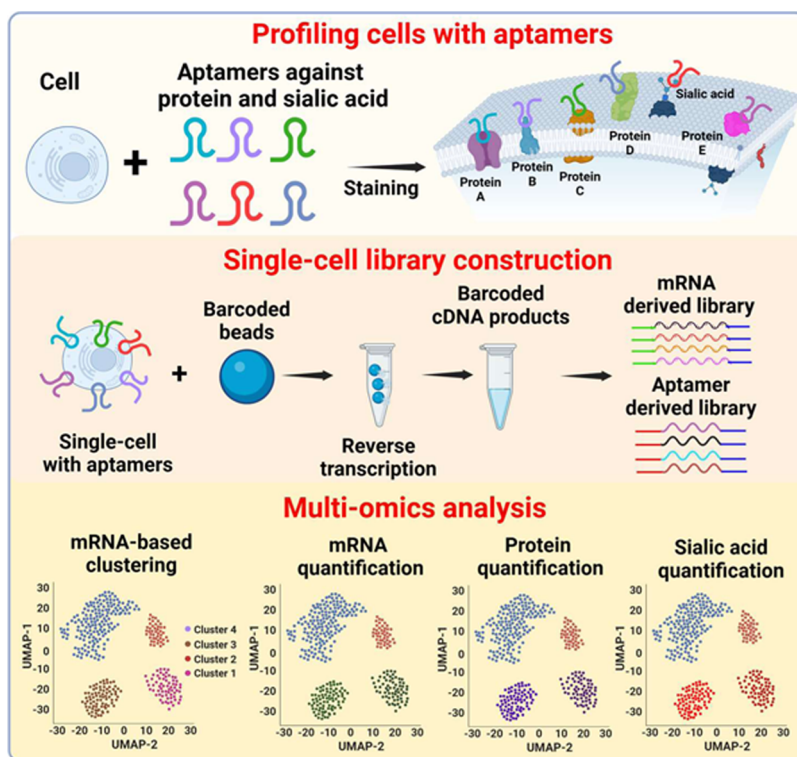


Figure 1. Scheme of Apt-seq workflow for single-cell multiomics analysis. Aptamers were designed applicable for the 10× Genomics single-cell RNA-seq platform. After incubating cells with aptamers, mRNA and aptamer libraries were constructed following the manufacturer's protocol, enabling integration of cell surface protein and glycan data with single-cell transcriptomes for interdependent analysis. Subsequently, the multiomics data were analyzed to gain comprehensive insights into the cellular landscape, combining information from mRNA, protein, and glycan.

surface proteins and mRNA in individual cells.^{12–14} However, antibody is costly, and the synthesis of DNA-barcoded antibody conjugates is complex. In addition, the large molecular weight of antibodies may cause steric hindrance when multiple antibodies are used to bind multiple proteins simultaneously. More importantly, most antibodies are designed to recognize proteins, leaving small molecules, glycans, and non-proteins biomolecules largely undetectable. This limitation significantly constrains the capacity of antibody-seq to comprehensively profile the molecular diversity present in biological systems. For glycan profiling, there are currently sequencing-based methods that utilize lectin-DNA barcodes¹⁵ and chemical enzyme labeling with fucosyltransferase.¹⁰ However, these methods encounter technical challenges, including complex enzymatic labeling procedures, variations in labeling efficiency, and the potential disruption of fragile cellular states due to prolonged treatments, all of which may affect the accuracy and reliability of glycan profiling results. These drawbacks call for the development of simple and efficient molecular tools that facilitate high-throughput multimodal molecules profiling and quantification at single-cell resolution, while remaining accessible and affordable.

Aptamers, often termed as “chemical antibodies,” are short single-stranded oligonucleotides capable of specific binding to nonsequencable target molecules, including proteins and small molecules.^{16–18} Unlike antibodies, which necessitate laborious and costly *in vivo* screening, aptamers can be rapidly and cost-effectively generated in large quantities through *in vitro* selection within days using Systematic Evolution of Ligands by Exponential Enrichment (SELEX).^{19,20} Furthermore, the relatively small size of aptamers (<30 kDa) in comparison to antibodies (~150 kDa) confers a distinct advantage in their

ability to access the cell surfaceome. In addition, aptamers can be selected through live cell-based approaches and synthesized chemically with high reproducibility, effectively eliminating issues associated with batch-to-batch variability. As nucleotide acids, aptamers can be directly sequenced without additional DNA barcode conjugation.^{21–24} These unique features making them ideal tools for high-throughput mapping of diverse molecular entities in the quantitative science.

In this study, we extend Apt-seq for the simultaneous quantification of multimodal omics—including mRNA, cell surface proteins and, glycans—in both cell lines and complex clinical samples, thereby establishing Aptomics as a new modality that leverages aptamers for integrated, multiplexed molecular profiling at the single-cell level. In contrast to Delley et al.²⁵ who extended a poly-A tail at the 3' terminus of aptamers (potentially interfering with transcriptome capture), we employ a capture sequence that enables independent, noninterfering capture of both aptamers and mRNA. Additionally, we have expanded the aptamer target panel to include not only cell surface proteins but also cell surface glycans and metabolites, thereby providing a more comprehensive analysis of cellular composition. When applied to clinical samples, our approach introduces new insights into tumor heterogeneity and the differentiation of T cells, thereby broadening the scope of potential applications. We initially validated the feasibility of our approach at both bulk and single-cell levels using commercial cell lines, comparing its performance with flow cytometry data. When applied to complex clinical samples, Apt-seq facilitated the investigation of tumor heterogeneity at the protein level, identifying a subset of tumor cells exhibiting elevated surface expression of PTK7, a protein marker associated with enhanced cellular stemness. Additionally,

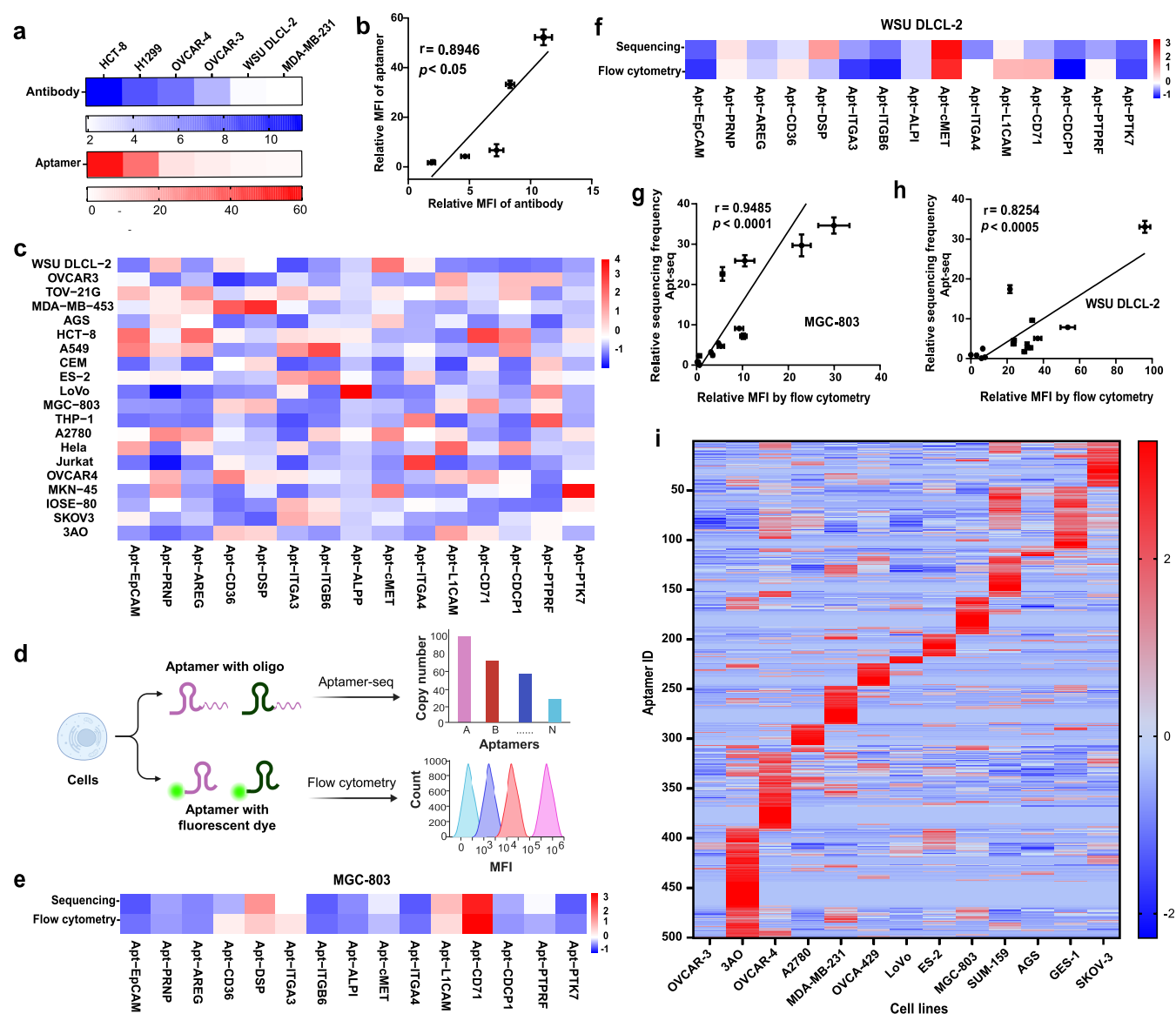


Figure 2. Apt-seq qualitatively characterizes cell surface proteins at bulk level. (a) Heatmap depicting the average relative log fluorescence intensities of aptamer and antibody binding to six indicated cell lines. Cells were incubated with either an aptamer targeting CD71 or an anti-CD71 antibody. Data represent the mean of three independent experiments. (b) Scatter plot showing the correlation between aptamer and antibody binding across the six cell lines ($n = 3$). (c) Apt-seq was also used to evaluate the binding pattern of 15 aptamers on 20 kinds of cell lines, indicating that Apt-seq could profile the cell surfaceome in a high-throughput manner. (d) Workflow to compare aptamer sequencing data with flow cytometry, the gold standard technique. (e) The quantitative correlation between Apt-seq and flow cytometry for MGC-803 cell line. (f) The quantitative correlation between Apt-seq and flow cytometry for WSU-DLCL-2 cell line. (g) The correlation between Apt-seq data and flow cytometry data for MGC-803 cell line was found to be significant according to Spearman's rank correlation coefficient (p). (h) The correlation between Apt-seq data and flow cytometry data for WSU-DLCL-2 cell line was found to be significant according to Spearman's rank correlation coefficient (p). (i) Binding patterns of an enriched pool across 13 different cell lines. Each cell line exhibits a distinct binding pattern when using 500 aptamers, demonstrating the ability of Apt-seq to differentiate among multiple cell lines. "Aptamer ID" refers to the aptamer identification code, with each line representing a distinct sequence. Each bar represents the mean of technical triplicates.

using an aptamer targeting sialic acid, we achieved simultaneous detection of mRNA, cell surface proteins, and cell surface glycans. By analyzing the dynamic changes in sialylation during T cell differentiation, we observed an increase in sialic acid levels as resting T cells transitioned into functional T cells, followed by a subsequent decline upon maturity. Taken together, Apt-seq enables the simultaneous quantitation of both directly and nondirectly sequencable molecules, establishing Aptomics as a versatile and scalable platform for the comprehensive quantification of molecules of

life in complex biological systems and advancing quantitative science.

RESULTS AND DISCUSSION

Apt-seq Workflow in Single-Cell Multiomics. In this study, we present Apt-seq, a high-throughput quantitative analysis of biomolecules including mRNA, proteins and glycans at single-cell resolution (Figure 1). Leveraging the 10× Genomics single-cell (sc)RNA-seq platform,²⁶ which utilizes a droplet-based system designed for 3'-end mRNA digital counting across thousands of single cells, we extended

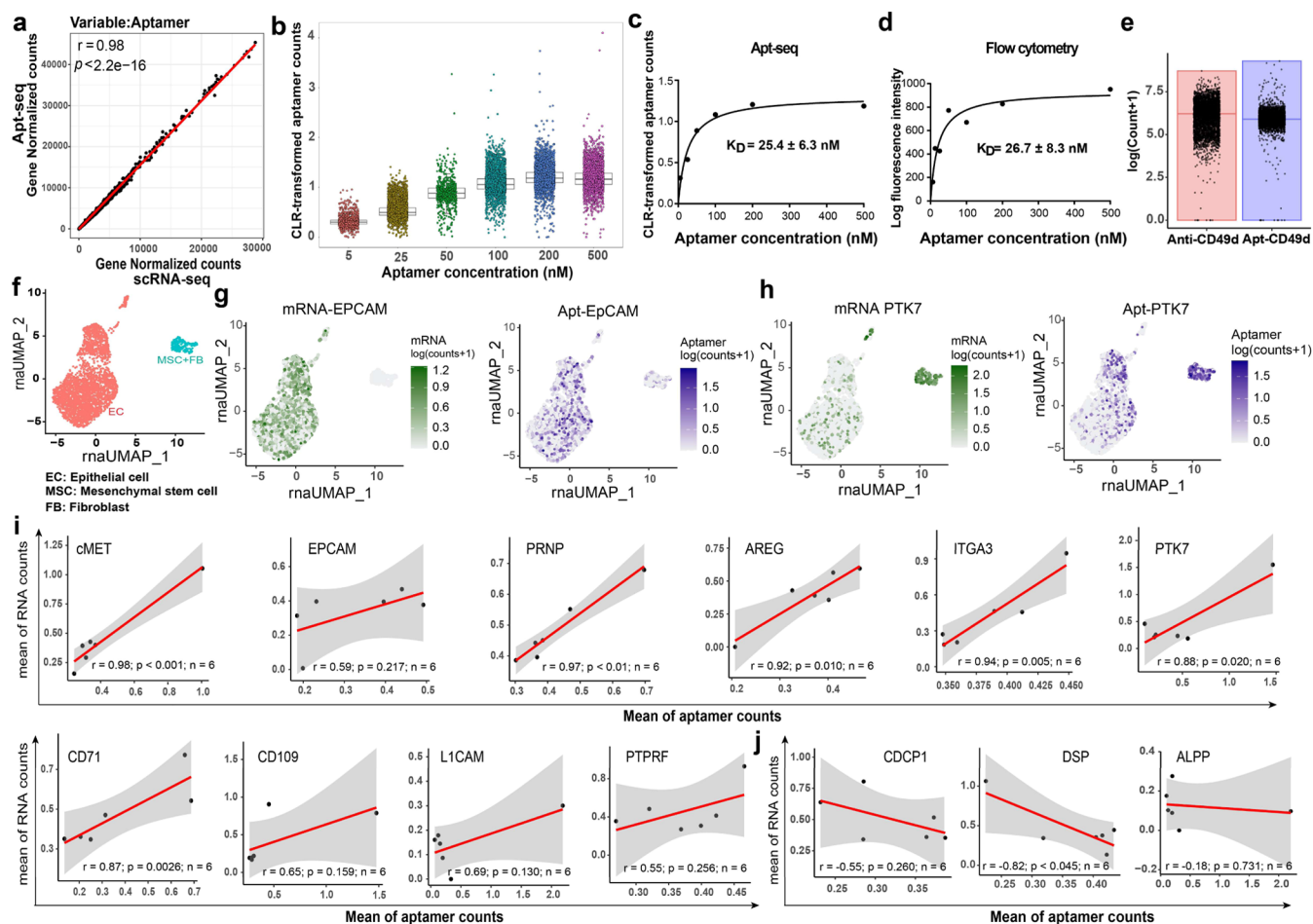


Figure 3. Apt-seq qualitatively characterizes cell surface proteins at the single-cell level. (a) CCRF-CEM cells (incubated with or without aptamer) were sequenced, and their transcriptomes were analyzed. A comparison of the top 4000 most highly expressed genes between two groups showed a strong correlation ($r = 0.98$, $p < 2.2 \times 10^{-16}$), indicating that the aptamer has no influence on the transcriptomes. (b) Box plot showing apt-PTK7 counts using series of concentration measured by Apt-seq (c) Apparent equilibrium dissociation constant (K_D) measured using Apt-seq data. (d) Apparent equilibrium dissociation constant (K_D) measured by flow cytometry. (e) Comparison of single-cell sequencing data between Apt-seq and antibody-seq, showing comparable read counts. (f) Transcriptome-based clustering of single-cell expression profiles obtained from ovarian cancer tissue by Apt-seq reveals distinct cell populations, including epithelial cells (ECs), mesenchymal stem cells (MSCs), and fibroblasts (FBs). (g) UMAP projection of mRNA (green) and aptamer (blue) signals for EpCAM, showing predominant expression on ECs. (h) UMAP projection of mRNA (green) and aptamer (blue) signals for PTK7, showing predominant expression on MSCs and FBs. (i) Most aptamers (10 of 13) showed positive correlation between normalized mRNA expression and CLR-transformed aptamer counts at the cluster level. (j) Three targets showed negative correlation between normalized mRNA expression and CLR-transformed aptamer counts at the cluster level.

aptamers with a universal PCR handle and a sequence complementary to the capture sequence on single-cell beads (Table S1) to enable seamless integration with this platform. This design ensures that aptamers and mRNA are captured independently, preventing mutual interference and preserving the accuracy of sequencing data (Figure S1). To evaluate whether these extensions impacted the binding specificity of the aptamers to their respective targets, we selected Apt-CDCP1 (selected from an adherent cell line) and Apt-CD62L (selected from a suspension cell line) for proof-of-concept analysis. DNA secondary structures of the aptamer were predicted using DNAFold revealing that the extended aptamers version retained similar secondary structures to their original counterparts (Figure S2a). Furthermore, flow cytometry based binding assays confirmed that these extensions did not compromise their binding capacities of both aptamers (Figure S2b). To further validate these findings, we performed flow cytometric analysis on all 17 aptamers,

demonstrating that the addition of extension sequences did not influence their binding capabilities (Figure S3).

After incubating aptamers with single-cell suspensions, Chromium Next GEM Single Cell 3' Reagent Kits v3.1 (dual index) was used for constructing mRNA expression and aptamer libraries. This approach allows the independent capture and amplification of cellular mRNA and surface-bound aptamers via distinct sequences, effectively preventing cross-contamination between aptamer and mRNA libraries during sequencing. Additionally, it ensures sufficient sequencing depth for both mRNA and aptamers. After sequencing, we performed bioinformatics analysis by initially clustering cells based on their mRNA expression profiles to identify distinct cellular populations. Subsequently, we mapped each target's mRNA, protein, and sialylation levels onto these clusters. The integration of these multidimensional data allows for a comprehensive understanding of cellular heterogeneity and provides insights into the molecular characteristics of each cluster.

Apt-seq Profiling of Cell Surface Proteins at the Bulk Level. To demonstrate aptamers showed similar binding affinities with antibodies, we first compared the binding trends of an aptamer targeting CD71²⁷ and anti-CD71 antibody in six kinds of cells. We observed a strong positive correlation ($r = 0.8946$, $p < 0.05$) between aptamer and antibody (Figure 2a,2b), highlighting the potential of aptamers as a promising alternative to antibodies for profiling cell surface proteins. Furthermore, to evaluate whether Apt-seq can quantitatively measure the expression of cell surface proteins in a high-throughput way, we performed profiling using a panel of well-characterized aptamers against 15 distinct cell surface proteins across 20 cell lines (Figure 2c). The precise quantitative sequencing was achieved by incorporating a unique molecular identifiers (UMI) as described in our previous work.²⁸ The UMI library sequences are listed in Table S2. To validate the accuracy of Apt-seq, we benchmarked the sequencing copy number of aptamers with the fluorescence intensity of the gold standard technique flow cytometry (Figure 2d). We selected two representative cell lines—a suspension cell line (WSU DLCL-2) and an adherent cell line (MGC-803)—for flow cytometry-based comparison. The results demonstrated a strong positive correlation between sequencing and flow cytometry data for both cell lines: MGC-803 ($r = 0.9485$, $p < 0.0001$; Figure 2e,2g) and WSU DLCL-2 ($r = 0.8254$, $p < 0.0005$; Figure 2f,2h). These findings demonstrate that Apt-seq is a robust and high-throughput method for digitizing cell surface molecules.

The aptamer screening enrichment libraries contain tens to hundreds of thousands of sequence variants, encompassing a diverse repertoire of potential aptamers targeting various molecules. We next sought to assess the capability of aptamer-enriched libraries as probe pools for higher-throughput, multidimensional, and multiparametric profiling of cell surface differences. Initially, we performed 16 rounds of cell-SELEX using ovarian cancer cell line OVCAR-4 as the positive target, yielding a highly enriched aptamer pool (R16). Flow cytometry analysis demonstrated that R16 exhibited strong binding to OVCAR-4 cells and varied binding ability across 12 other cell lines (Figure S4), suggesting the potential of this aptamer library to distinguish between different cell lines. Flow cytometry has limitations, as it only provides average binding intensity and does not offer information on the sequence-specific binding of each aptamer. To address this, we subjected the R16 library to next-generation sequencing (NGS) after incubation with various cell lines. The coefficient of variation (CV) from four independent sequencing experiments was less than 15% (Figure S5a), and the Spearman correlation coefficients were above 0.97 (Figure S5b and 5c), indicating high stability and reliability of aptamer sequencing. Further aptamer clustering analysis revealed 500 aptamers within the R16 library capable of distinguishing 13 cell lines based on their unique binding patterns (Figure 2i), demonstrating the potential of Apt-seq for multiparameter cell characterization. The sequences of all 500 aptamers are summarized in Table S3. Notably, although both OVCAR-3 and OVCAR-4 are high-grade serous ovarian cancer cell lines, R16 bound significantly weaker to OVCAR-3 compared to OVCAR-4. This observation aligns with known genomic and proteomic differences between these two cell lines,^{29–31} further validating the efficacy of aptamers in recognizing cell surface molecular heterogeneity.

Apt-seq Profiling of Cell Surface Proteins at the Single-Cell Level. Simultaneous detection of multiomics at the single-cell level is crucial for identifying cell types, dissecting cellular heterogeneity, and elucidating the functions of biomolecules. Hence, to test the applicability of Apt-seq at single-cell resolution, we first assessed whether the incubation of aptamers had an influence on the transcriptional status of cells, we conducted single-cell sequencing on cells with and without aptamer incubation. The high correlation ($r = 0.98$, $p < 2.2 \times 10^{-16}$) of gene expression between the two samples demonstrates that the introduction of aptamer treatments had no discernible effect on the transcriptional profiles of cells (Figure 3a). Next, to further demonstrate the quantitative capabilities of Apt-seq in characterizing cell surface proteins at the single-cell level, a titration assay with six serial concentrations (5 nM, 25 nM, 50 nM, 100 nM, 200 nM, and 500 nM) of apt-PTK7 was conducted on CCRF-CEM cells. Analysis of single-cell sequencing data revealed a dose-dependent binding of apt-PTK7 with CCRF-CEM cells, ultimately reaching a plateau at approximately 200 nM (Figure 3b). The apparent equilibrium dissociation constant (K_D) measured through single-cell sequencing closely matched the K_D determined using flow cytometry (25.4 nM vs 26.7 nM) (Figure 3c,3d). These results underscore the potential of Apt-seq for quantitative profiling of molecules at the single-cell level. In addition, to demonstrate the robustness of Apt-seq at single-cell level, we did comparative analysis with CITE-seq, which uses antibody-oligo complexes as quantitative probes for cell surface proteins.¹² Single-cell sequencing results showed that reads obtained using the CD49-targeting aptamer were comparable to those obtained with antibody-oligo complexes, further demonstrating the accuracy and sensitivity of Apt-seq (Figure 3e).

Subsequently, we employed Apt-seq to analyze highly heterogeneous clinical solid cancer tissue samples. Aptamers were incubated with single cells isolated from ovarian cancer tissue, followed by Apt-seq analysis. Uniform manifold approximation and projection (UMAP) of transcriptomes revealed the presence of epithelial tumor cells, mesenchymal stem cells, and fibroblasts within the sample (Figure 3f). We found that the aptamer targeting EpCAM, a protein marker for epithelial cells, predominantly localized to epithelial cells, and its abundance significantly correlated with EPCAM mRNA levels (Figure 3g,3i). Similarly, the aptamer targeting PTK7 protein exhibited significant correlation with PTK7 mRNA levels (Figure 3h,3i) and was mainly distributed among mesenchymal stem cells and fibroblasts, consistent with PTK7 expression in these cell types.^{32,33} Overall, ten out of 13 aptamers demonstrated significant positive correlations between the abundance of their target proteins and corresponding mRNA levels (Figure 3i). In contrast, three aptamers (targeting CDCP1, DSP, and ALPP) did not show such correlations (Figure 3j), likely due to discrepancies between mRNA expression and surface protein levels. These findings further validate the specificity and accuracy of Apt-seq in complex solid cancer tissue samples, and emphasize the importance of directly quantifying proteins.

Apt-seq Enables Sensitive Profiling of Cell Heterogeneity in Clinical Blood Tumor Samples. We have validated the applicability of Apt-seq in clinical solid tumor samples. To further assess the feasibility of Apt-seq in clinical blood samples, we utilized a bone marrow sample from a newly diagnosed multiple myeloma (MM) patient. The sample was

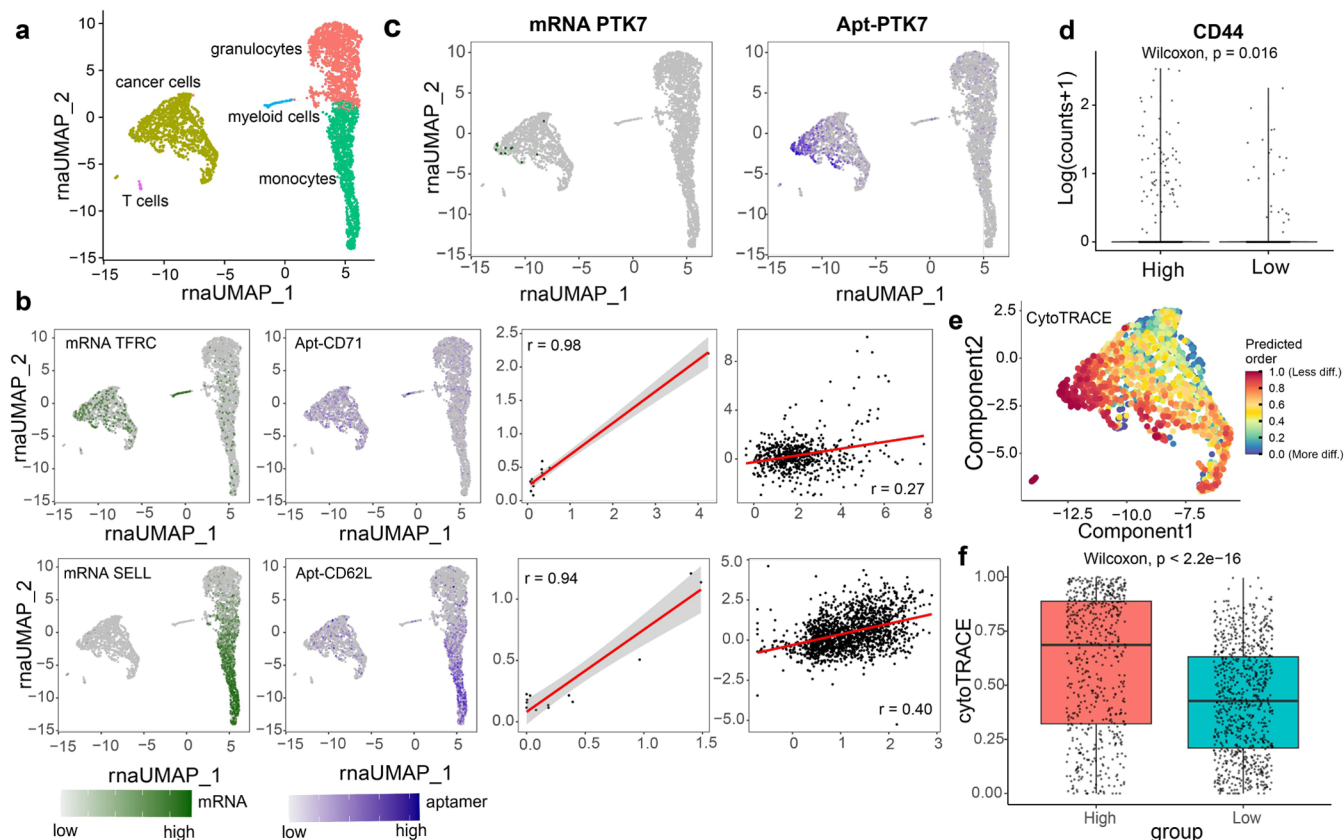


Figure 4. Apt-seq enables detailed characterization of tumor cell heterogeneity in clinical blood tumor samples. (a) Transcriptome-based clustering of Apt-seq single-cell expression profiles of multiple myeloma sample reveals distinct cell populations. (b) UMAP distribution of two representative genes and aptamers on the UMAP plot from panel (a). The mRNA was marked in green and corresponding aptamer was marked in blue. Dot plots reveal the correlation between aptamer counts and mRNA expression levels of corresponding targets per cell, as measured by single-cell sequencing. (c) UMAP distribution of PTK7. Not all cancer cells bound to apt-PTK7. (d) Violin plots of CD44 mRNA expression in Apt-PTK7 high-binding cells and low-binding cells. (e) Stemness of PTK7^{high} and PTK7^{low}-expressing cell groups are exhibited through CytoTRACE analysis. (f) Box plot of CytoTRACE scores for cell stemness between PTK7^{high} and PTK7^{low}-expressing cell groups.

incubated with aptamers and subsequently subjected to Apt-seq UMAP based on mRNA expression revealed five distinct cell populations, including granulocytes, myeloid cells, monocytes, T cells, and a substantial tumor cell population (Figures 4a and S6). Notably, we observed that the aptamer targeting CD71 predominantly localized to both tumor and myeloid cells, whereas the aptamer targeting SELL protein was primarily distributed in monocytes (Figure 4b). The abundance of these aptamers showed a significant correlation with the mRNA expression levels of their respective targets, underscoring the robustness and precision of Apt-seq as a platform for molecular profiling.

Notably, we observed that aptamers exhibited superior sensitivity in detecting cellular markers compared to mRNA sequencing, particularly for low-expressed mRNAs. For instance, in the case of the PRNP aptamer, which binds to prion proteins, only a small proportion of cancer cells expressed detectable levels of PRNP mRNA. In contrast, the distribution of the PRNP aptamer was significantly broader among these cells (Figure S7), further supporting findings from antibody-seq studies.^{12,13} This suggests that Apt-seq facilitates more sensitive detection of cellular phenotypes. Further analysis identified a subset of tumor cells in MM samples that exhibited strong binding to the PTK7 aptamer (Figure 4c). This indicates that these tumor cells express elevated levels of PTK7, in contrast to other tumor cells,

thereby highlighting the heterogeneous nature of the tumor cell population. To further investigate the transcriptional features associated with cells exhibiting high and low PTK7 aptamer binding, we performed differential gene expression analysis between PTK7^{high} and PTK7^{low} tumor cells. Among the top ten differentially expressed genes between these two subgroups, CD44, a stemness marker, showed significantly higher expression in PTK7^{high} cells compared to PTK7^{low} cells (Figures 4d, S8, and S9). CytoTRACE analysis indicated that PTK7^{high} tumor cells displayed higher stemness than PTK7^{low} tumor cells, which is consistent with literature suggesting the involvement of PTK7 in regulating tumor stemness³⁴ (Figures 4e, 4f). However, owing to the relatively low sensitivity of mRNA detection, distinguishing between these two tumor cell types based on PTK7 mRNA expression is challenging (Figure 4c). These findings highlight the potential of integrated multimomics to uncover tumor cell heterogeneity and characterize cellular phenotypes.

Multimodal Profiling of the Transcriptome, Cell Surface Protein, and Cell Surface Metabolite in Clinical Blood Sample. In comparison to antibodies, aptamers exhibit a broader capacity for target recognition. Consequently, Apt-seq is well-suited for simultaneous multimodal profiling, encompassing transcriptome, cell surface protein, and cell surface metabolite detection. To demonstrate this, we incorporated an aptamer targeting sialic acid into our aptamer

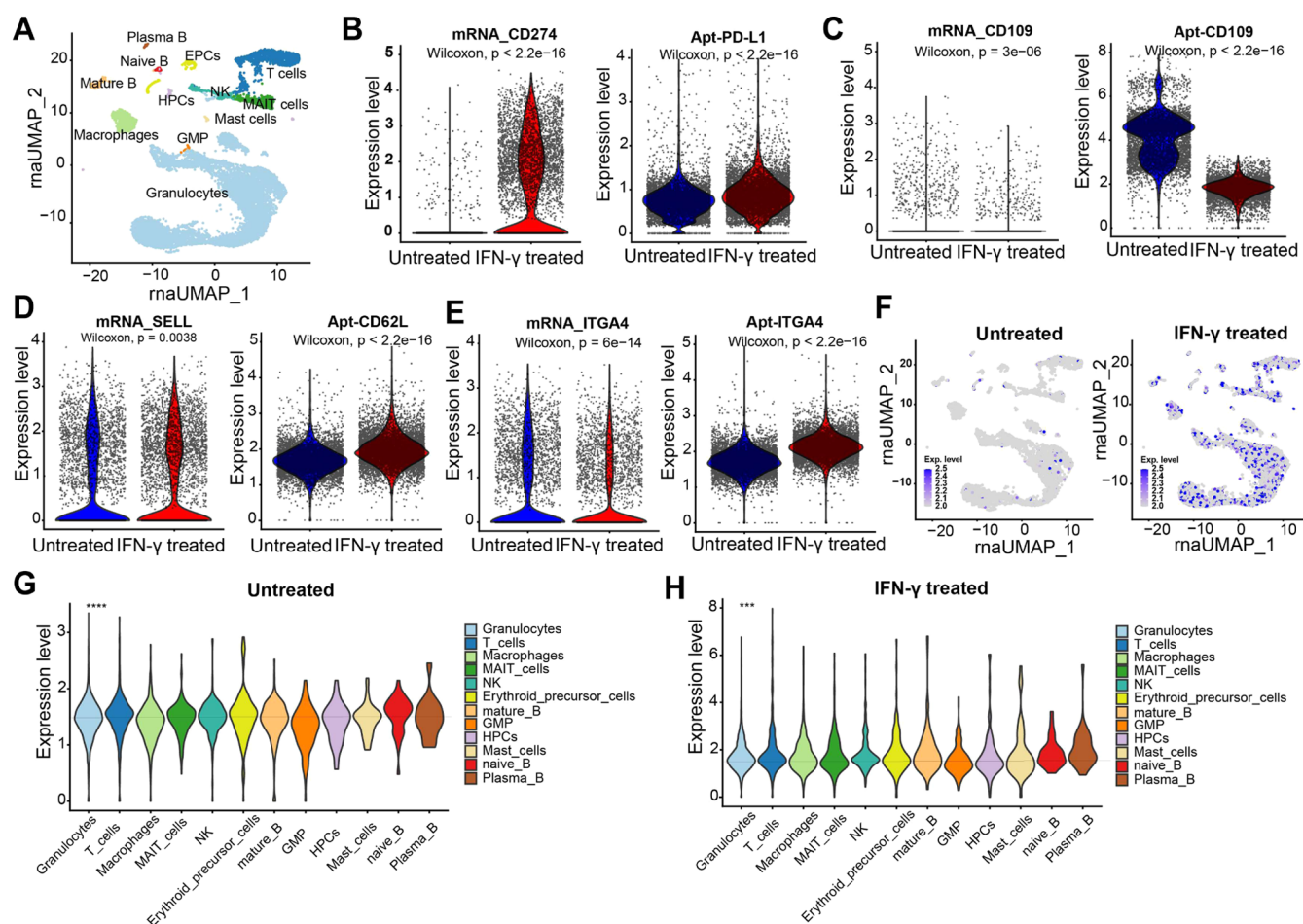


Figure 5. Multimodal profiling of the transcriptome, protein, and sialic acid in clinical blood sample. (a) UMAP visualization of transcriptomebased clustering from Apt-seq single-cell expression pro-files of multiple myeloma sample. (b–e) Violin plots comparing mRNA expression and aptamer binding for CD274 (b), CD109 (c), SELL (d), and ITGA4 (e). (f) UMAP visualization showing sialic acid levels based on aptamer binding in control samples (left) and IFN- γ -treated samples (right). (g, h) Violin plot showing sialic acid levels across distinct cell clusters in untreated samples (g) and IFN- γ -treated (h) samples. *** $p < 0.001$, **** $p < 0.0001$.

panel^{9,35} and evaluated its binding to cells treated with or without $\alpha 2$ –3,6,8,9-neuraminidase A. A nonaptamer randomized sequence of the same length was used as a negative control. The results demonstrated that robust binding of Apt-sialic acid, which was significantly reduced upon $\alpha 2$ –3,6,8,9-neuraminidase A treatment, confirming the aptamer's specificity (Figure S10).

Next, we performed Apt-seq using multiple myeloma bone marrow samples treated with or without interferon- γ (IFN- γ), which is a key cytokine that activates and regulates monocyte-macrophage differentiation and function. Transcriptome-based clustering reveals distinct cell populations, including B cells, T cells, macrophages, natural killer cells, endothelial progenitor cells, hematopoietic progenitor cells, mast cells, granulocyte-monocyte progenitors (GMPs) and a substantial population of granulocytes (Figure 5a). And the IFN- γ treatment group had a similar cell type composition (Figure 5f, right). Interestingly, the number of aptamers detected per cell was relatively higher than the number of mRNAs, likely due to the high diversity of mRNA molecules across different genes, which may lead to reduced sensitivity in detecting less abundant mRNAs. In contrast, the aptamer approach measures a smaller set of proteins, potentially resulting in more consistent and detectable signals. Additionally, while mRNA expression

often correlates with protein abundance, this relationship is not always direct due to regulatory factors such as translation efficiency, protein stability, post-translational modifications, protein localization, and intracellular trafficking.^{29,30} Further bioinformatics analysis revealed a significant upregulation of both PD-L1 mRNA and its corresponding aptamer levels following IFN- γ treatment, which is consistent with previous reports,³⁶ validating the reliability and success of our experimental approach (Figure 5b). For genes, such as CD109 and SELL, the changes in mRNA and its cell surface protein expression is consistent (Figure 5c,d). However, for other genes, like ITGA4, we observed discrepancies between mRNA and protein expression (Figure 5e). This finding highlights the spatiotemporal variation between mRNA and protein expression and underscores the importance of directly characterizing protein levels to achieve a comprehensive understanding of cellular states.

Interestingly, mapping sialic acid on the UMAP plot revealed that IFN- γ treatment significantly enhanced the overall binding capacity of the sialic acid aptamer (Figure 5f). To validate this observation, we used the B-lymphoma cell line Ramos to compare the trends in binding changes observed for the sialic acid aptamer with those of lectins. Consistently, IFN- γ treatment enhanced the binding capacity of both the

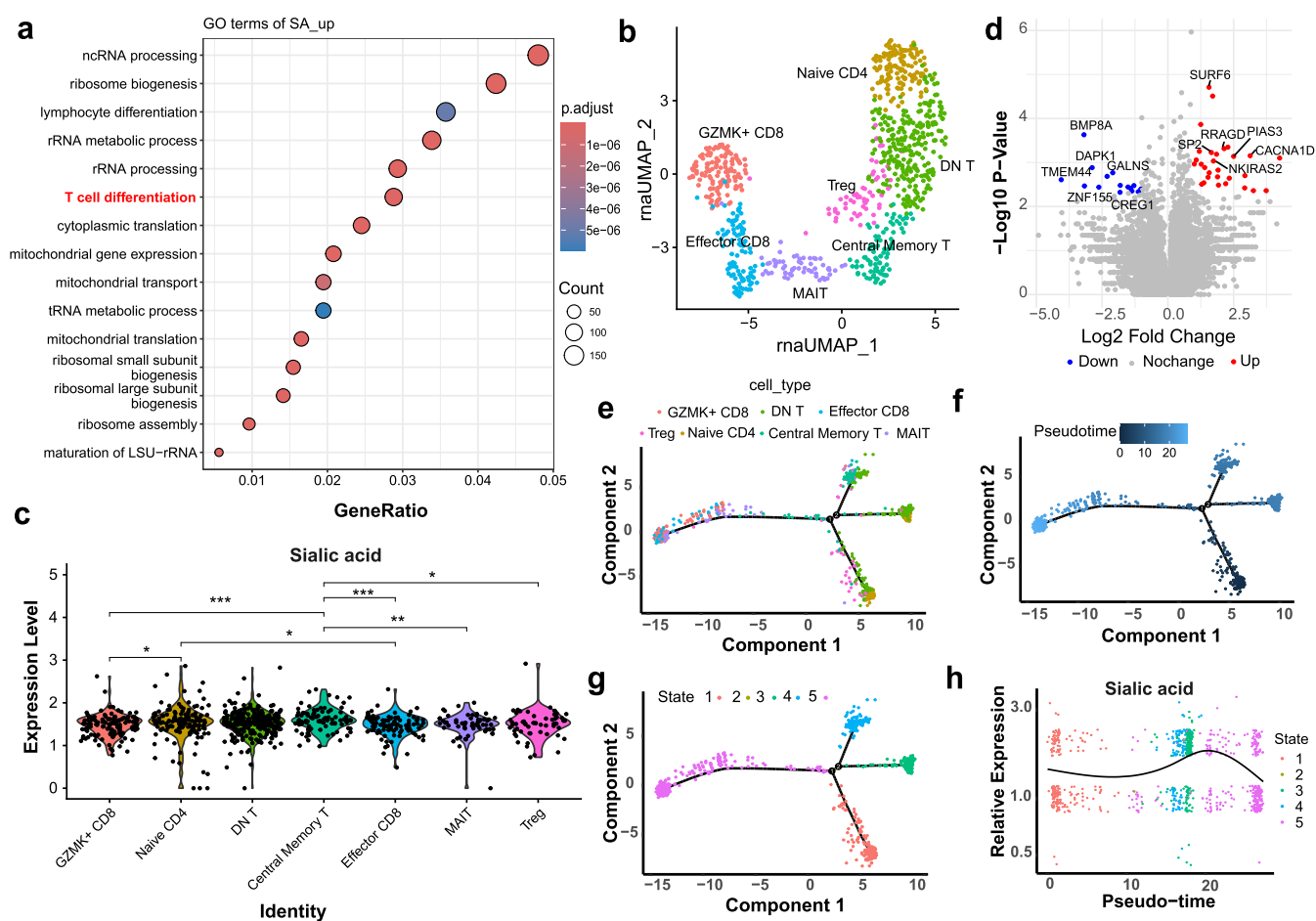


Figure 6. Track changes in cell surface metabolite levels during T cell differentiation using Apt-seq (a) Gene Ontology (GO) analysis revealing biological process enrichment of upregulated transcripts in "SA_{high}" versus "SA_{low}" groups in untreated MM bone marrow cells. (b) UMAP visualization displaying sialic acid levels based on aptamer binding in T cells from untreated multiple myeloma bone marrow sample. (c) Violin plot depicting sialic acid levels across T cell activation subsets. (d) Volcano plot showing transcriptional differences between "SA_{high}" and "SA_{low}" T cells. (e-f) Monocle plots depicting the pseudotime ordering and differentiation trajectory of T cells in untreated MM bone marrow samples. (g-h) Monocle plots illustrating changes in sialic acid levels along the trajectory timeline. Numbers in the (e-h) plots represent nodes identified in the trajectory analysis, marking distinct cellular states, with each branch point indicating potential decision points in cell biological processes. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

sialic acid aptamer and lectins, corroborating the sequencing results (Figure S11). Further analysis of differences in sialic acid levels across cell types demonstrated that, irrespective of treatment, B cells exhibited the highest levels of sialic acid aptamer binding (Figure 5g,h). These findings suggest that Apt-seq, with its ability to target a wide range of entities, holds significant potential for the precise digital quantification of biomolecules that cannot be sequenced.

Track Changes in Cell Surface Metabolite Levels during T Cell Differentiation Using Apt-seq. Previous studies have established that T-cell activation and differentiation are associated with significant alterations in cell-surface receptor glycosylation patterns.^{10,37} In this study, we classified untreated multiple myeloma bone marrow cells into "SA_{high}" and "SA_{low}" groups based on sialic acid aptamer binding counts, using the median value as a threshold. Differentially expressed genes (DEGs) and functional enrichment revealed a correlation between sialic acid levels and T-cell differentiation (Figure 6a). We also assessed sialic acid levels in IFN- γ -treated T cells, but observed no significant differences across differentiation stages (Figure S12). As a result, we focused our analysis on the untreated samples.

Subsequently, we isolated T cells from these samples to compare sialic acid levels in T-cell subsets. The T cells were categorized into seven subsets according to their activation stages (Figure 6b). Notably, as T cells differentiated, we observed statistically significant differences in sialic acid levels across multiple subpopulations (Figure 6c). Further comparison of the transcript differences between the "SA_{high}" and "SA_{low}" groups revealed that genes associated with T-cell activity, including PIAS3, NKIRAS2, CACNA1D, RRAGD, SP2, and SURF6, were significantly elevated in the "SA_{high}" T cell population (Figure 6d).

To track dynamic changes in sialic acid level during T cell differentiation, we employed the Monocle algorithm to analyze single-cell pseudotime and visualize sialic acid levels across T-cell subsets (Figure 6e,f). The level of sialic acid in pseudotime was found to increase during the transition of resting T cells into functional T cells and decrease after cellular maturity (Figure 6g,h). Taken together, these results suggest that Apt-seq is a robust technique for tracking multimodal molecules in complex samples. This method enables the simultaneous quantification and dynamic tracking of metabolite changes during T-cell differentiation, providing valuable

insights into the molecular mechanisms underlying immune cell activation and function.

CONCLUSIONS AND OUTLOOK

The human body is a molecular machine composed of a vast array of diverse molecules of life. Mapping the molecular atlas of the human body and characterizing key biomolecules enhance our understanding of human biology and health. High-throughput sequencing technologies have revolutionized data generation, enabling the development of multiomics fields such as genomics, transcriptomics, and epigenomics, all of which rely on direct sequencing. For molecules such as proteins and glycans that cannot be sequenced directly, innovative indirect sequencing approaches have been developed. For instance, antibody-seq technology allows for indirect quantification of corresponding protein targets by conjugating DNA oligos to antibodies.^{38,39} For metabolites such as glycans, Glycan-seq has been reported, utilizing chemo-enzymatic glycan labeling with DNA barcodes. However, the complexity of conjugation and generating antibodies for certain metabolites remain significant limitations.^{40,41} In the case of Glycan-seq, glycosyltransferases with flexible donor substrate scopes for incorporating probe molecules are limited, and enzymatic labeling may not be suitable for certain fragile samples.^{42–44} Therefore, there is an urgent need to develop multimodal, universal, and easy-to-operate high-throughput sequencing methods for simultaneously digitalizing molecules. Aptamers are sequenceable nucleic acids that exhibit high specificity for a broad range of targets, including proteins and metabolites. In this study, we developed an aptamer-based method, Apt-seq, which enables simultaneous molecular indexing of metabolites, proteins, and mRNA. This approach allows for comprehensive quantitative analysis of both sequenceable and unsequenceable molecules of life at the single-cell level in a high-throughput manner, providing a novel solution to current limitations in molecular profiling.

We first demonstrate that aptamers show similar binding specificity with antibodies, and further validate Apt-seq as a robust and accurate molecular profiling method by exhibiting its strong correlation of sequencing data with flow cytometry data at bulk level (Figure 2). To enhance the utility of Apt-seq, we also designed aptamers for the simultaneous characterization of transcriptomes, glycans, and proteins in clinical complex samples, ensuring compatibility with scRNA-seq platforms (Figure 3). This integration facilitates a deeper understanding of the molecular heterogeneity of cell populations and provides a comprehensive view of cellular physiology and pathology (Figure 4). Due to the spatiotemporal discrepancies in the expression of mRNA and proteins, the expression levels of mRNA and proteins, particularly membrane proteins, is not always corresponded significantly (Figure 5). This finding highlights the differences in time and space between mRNA and protein expression and underscores the importance of directly characterizing protein levels to achieve a comprehensive understanding of cellular states.

Glycans, which play pivotal roles in cellular communication, immune evasion, and disease progression, can be effectively profiled using Apt-seq by employing sialic acid aptamers. Apt-seq provides unique insights into the dynamics of glycan level during T cell differentiation. Specifically, sialic acid levels increase as resting T cells transition into functional T cells and decrease after they reach maturity (Figure 6). Collectively, the data demonstrates that Apt-seq could address significant

limitations in current proteomic and glycomic profiling methods, and represents a significant advancement in the field of single-cell molecular profiling, offering a scalable, versatile, and cost-effective approach for the comprehensive analysis of cell surface proteins, glycans and mRNA. Collectively, we present Apt-seq, a novel multiomics framework that employs aptamers as molecular translators to enable sequencing-based analysis of otherwise non-sequenceable biomolecules—such as proteins and glycans. Through the Aptomics strategy, both directly and indirectly sequenceable targets can be quantified in parallel, offering a scalable and flexible platform for comprehensive molecular profiling in complex biological systems and advancing the frontier of quantitative life sciences.

One promising direction for further advancement is the integration of Apt-seq with other cutting-edge single-cell omics technologies, such as ATAC combined with Select Antigen Profiling by sequencing (ASAP-seq),⁴⁵ single-nucleus chromatin accessibility and mRNA expression sequencing (SNARE-seq),⁴⁶ Perturb-seq⁴⁷ This integration would enable the creation of more comprehensive multiomics data sets, facilitating the simultaneous profiling of proteins, lipids, metabolites, and glycans. In addition, aptamers can be selectively engineered to target specific protein modifications, offering a unique advantage for probing subtle molecular changes.^{48–50} Such a holistic approach could greatly enhance our understanding of cellular functions and interactions across multiple parameters. In addition, the continuous development of novel computational techniques is expected to improve the resolution and analysis of these data sets, providing valuable insights into cell identity and function at the single-cell level. Despite the many advantages of Apt-seq and its compatibility with a variety of other omics technologies, there are still several challenges to overcome. A major concern is that most aptamers developed so far are focused on tumor biology, with relatively fewer targeting CD (cluster of differentiation) molecules in immune cells. Furthermore, aptamers for metabolites often show lower affinity, necessitating optimization for better performance. Moreover, previous studies have demonstrated that certain post-translational and post-transcriptional modifications of specific proteins or RNAs, rather than their overall abundance, are closely associated with disease progression.^{51,52} Expanding the repertoire of aptamers, particularly those recognizing post-translational modifications, and developing techniques for the simultaneous quantification of both protein expression and post-translational modifications at the single-cell level are key directions for our future research.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/jacs.5c01296>.

Experimental details, materials, and methods, including the buffers and oligonucleotides used, as well as the specifics of cell lines and culture conditions, descriptions of flow cytometry analysis for evaluating the binding of aptamers from the R16 library to cells, quantitative PCR, Apt-seq assays at both bulk and single-cell levels; profiling of cell lines using an enriched aptamer pool; preparation of single-cell samples from clinical specimens; statistical analyses, including single-cell RNA sequencing computational analyses, single-cell aptamer

sequencing computational analyses, pseudotime analysis, and gene ontology (GO) enrichment analysis, among others (PDF)

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Notes

The authors declare no competing financial interest.

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